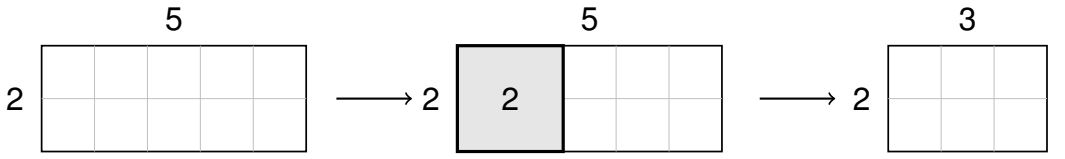
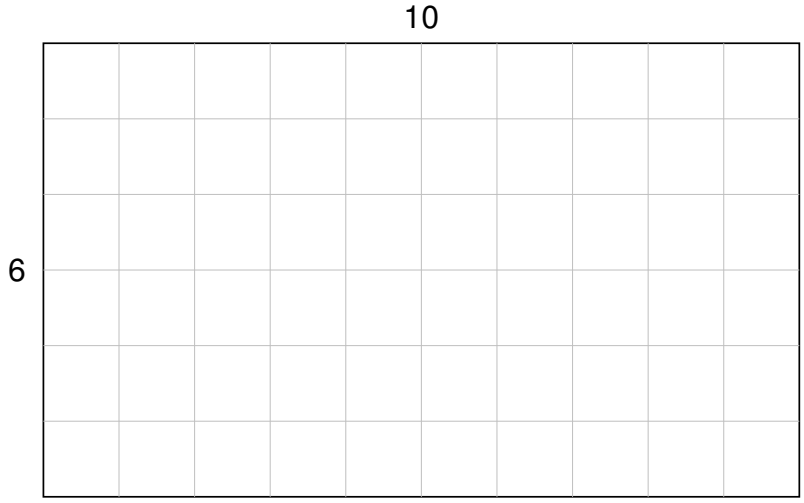
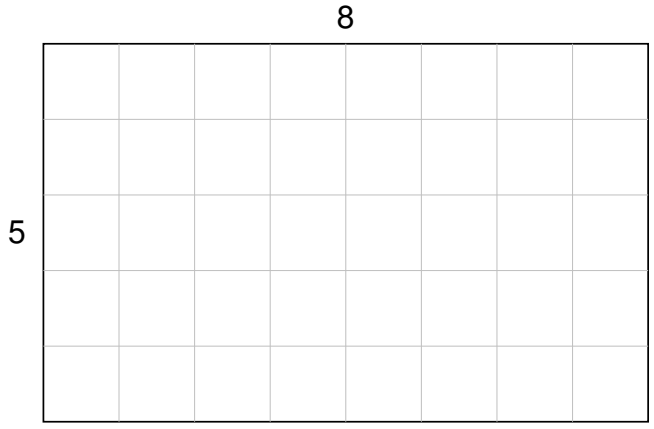
Keep square edges on grid lines. Cover rectangles with no gaps, overlaps, or overhang. For the biggest-square rule, cover the largest square at one end, then repeat on the rectangle left over.



Problem 1: Apply the biggest-square rule to both rectangles. Record every square size, including repeated sizes, and compare the last squares.

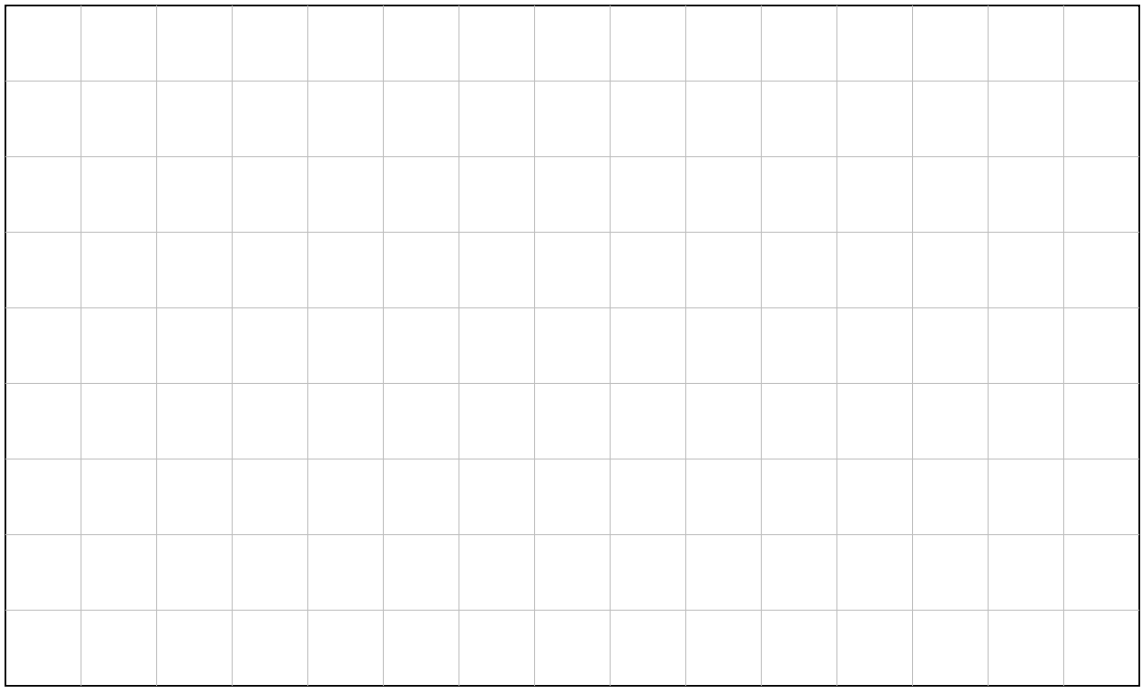




Problem 2: Find a rule that predicts the last square from the two starting side lengths.

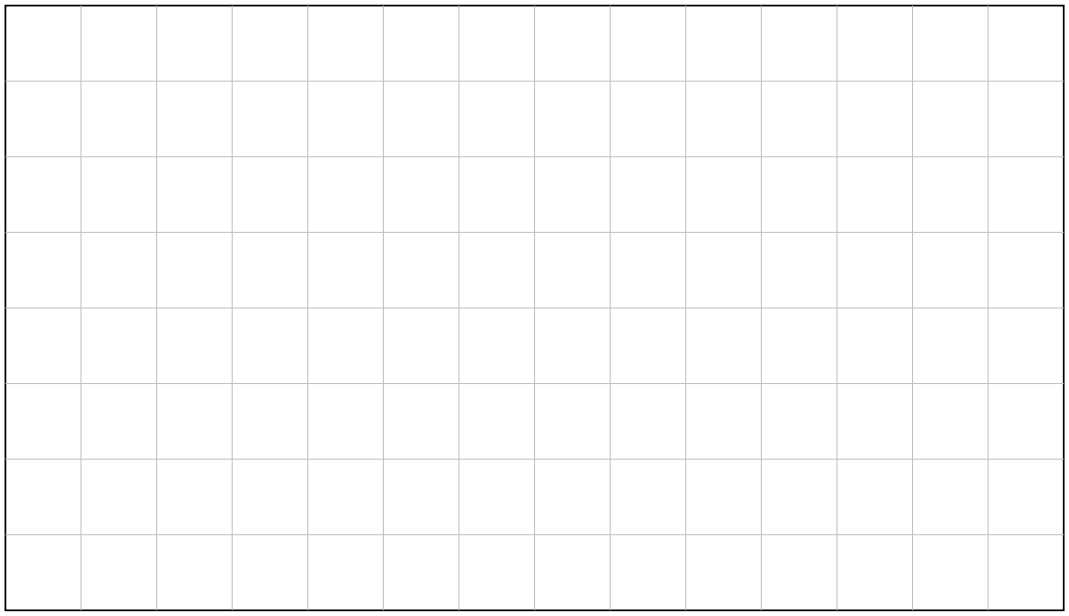
15

9

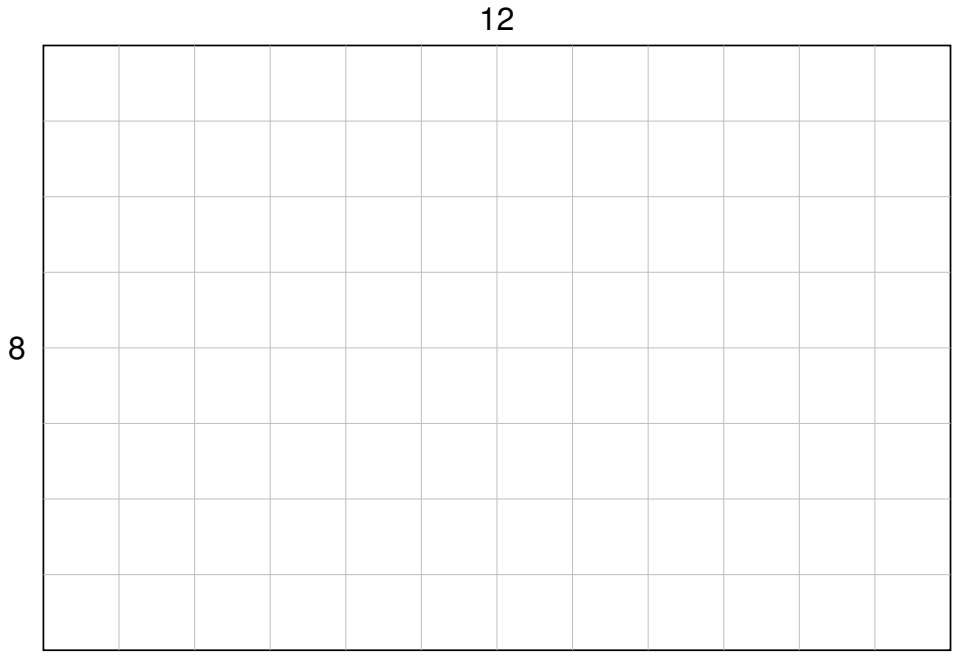


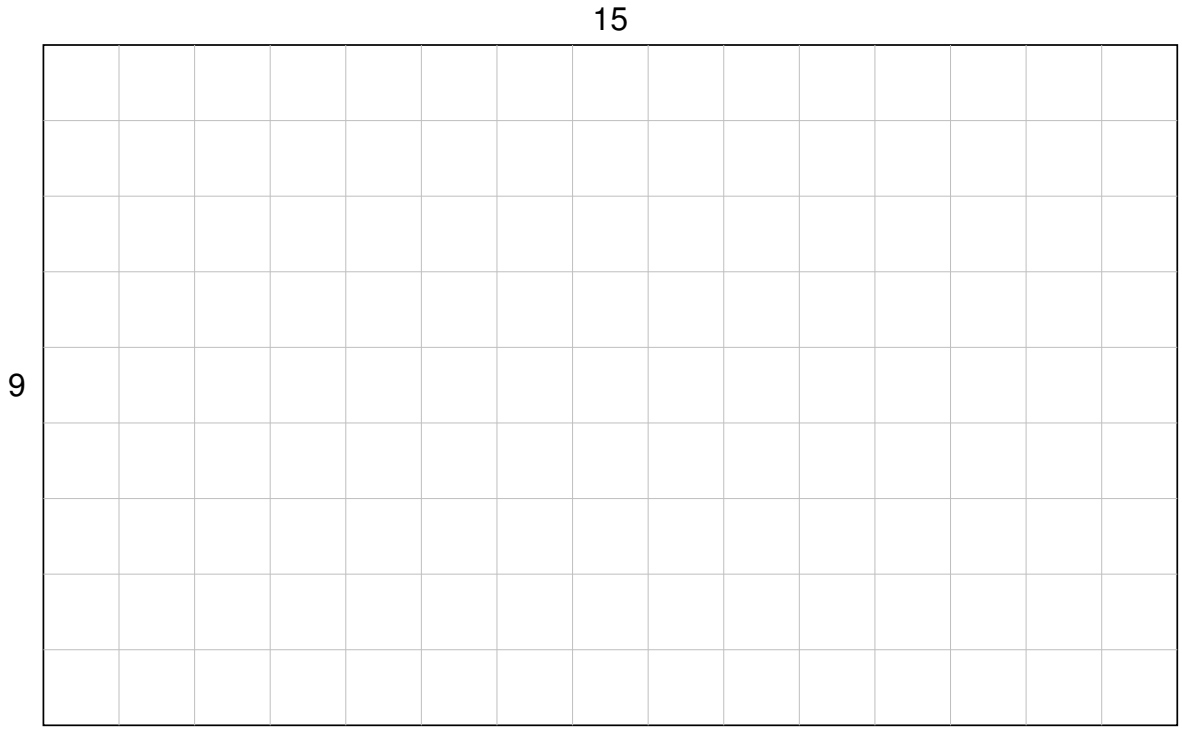
14

8

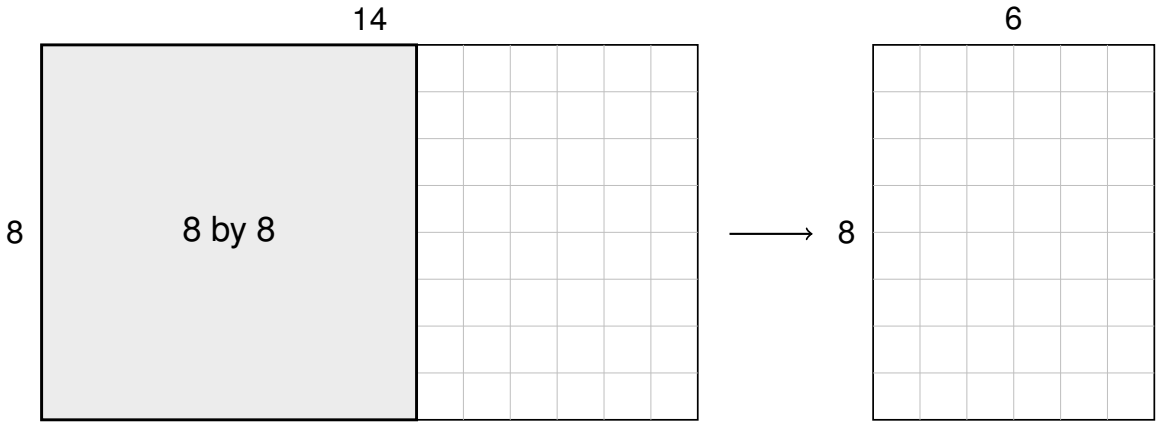


Problem 3: For each rectangle, find every size of identical whole-grid square that can cover it. Explain why your list is complete.





Problem 4: Which sizes of identical whole-grid square can cover the 14 by 8 rectangle? Which can cover the 6 by 8? Explain why these two lists always match after a biggest-square cut.



Problem 5: Explain why the last square from the biggest-square rule is the largest whole-grid square that can tile the starting rectangle using identical copies.

Problem 6: Choose a longer side from 9 through 15 and a shorter side from 2 through 8. Which rectangle produces the most different square sizes under the biggest-square rule? Does any other rectangle tie with it? How do you know?